Data Preprocessing & Exploratory Data Analysis

Dataset: Myntra T-Shirt Dataset (Kaggle)

1. Introduction

Objective

The objective of this project is to perform data preprocessing and exploratory data analysis on the Myntra T-Shirt Dataset obtained from Kaggle.

Dataset Description

The dataset contains information about T-shirt products available on Myntra.

Attributes:

- Brand
- Title
- Selling Price
- Price
- Discount

Number of Records: 4950

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2. Data Loading

Include:

import pandas as pd

df = pd.read_csv(...)

Add screenshot of:

df.head()

and

df.info()

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3. Data Preprocessing

Missing Values

df.isnull().sum()

Observation: No missing values found.

Missing Values:

Brand 0

Title 0

Selling Price 0

Price 0

Discount 0

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Duplicate Values

df.duplicated().sum()

df.drop_duplicates(inplace=True)

Observation: Duplicate records checked and removed.

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Data Type Conversion

df['Price']=pd.to_numeric(df['Price'])

df['Discount']=df['Discount'].str.replace('%','')

Observation: Converted text values into numerical format for analysis.

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4. Exploratory Data Analysis (EDA)

Summary Statistics

df.describe()

Screenshot

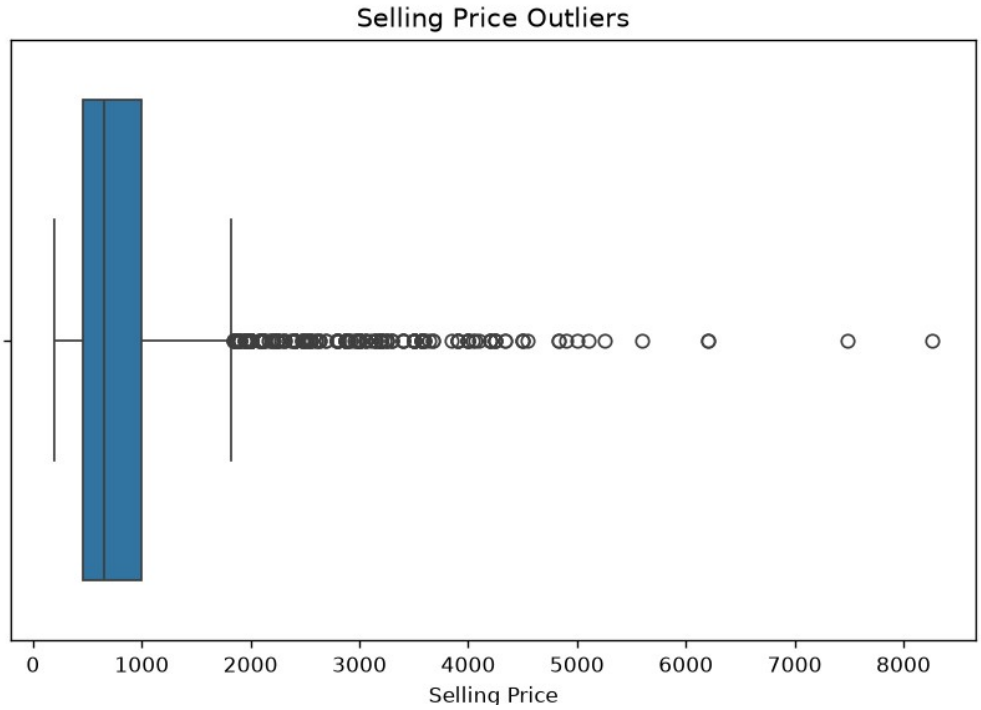
Summary Statistics:

	Selling Price	Price	Discount
count	2815.000000	2763.000000	2763.000000
mean	905.612789	1373.029316	35.779949
std	772.847844	994.062840	19.698342
min	199.000000	299.000000	0.000000
25%	454.000000	799.000000	21.000000
50%	649.000000	1099.000000	40.000000
75%	1000.000000	1499.000000	50.000000
max	8260.000000	11800.000000	77.000000

Observation

- Average selling price identified.
- Maximum and minimum prices observed.

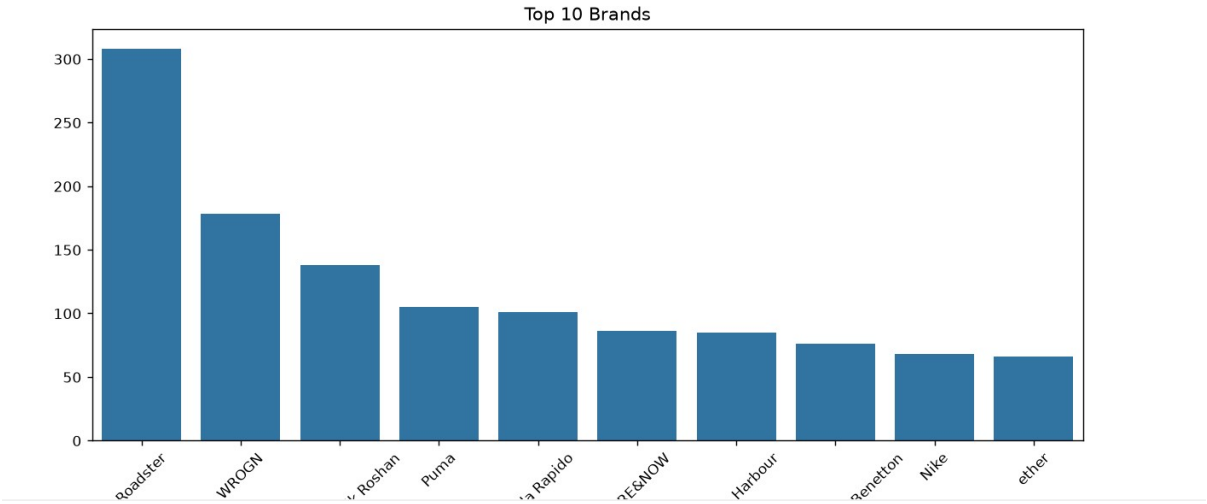
Boxplot of Selling Price



Observation

Some products have exceptionally high prices, indicating outliers.

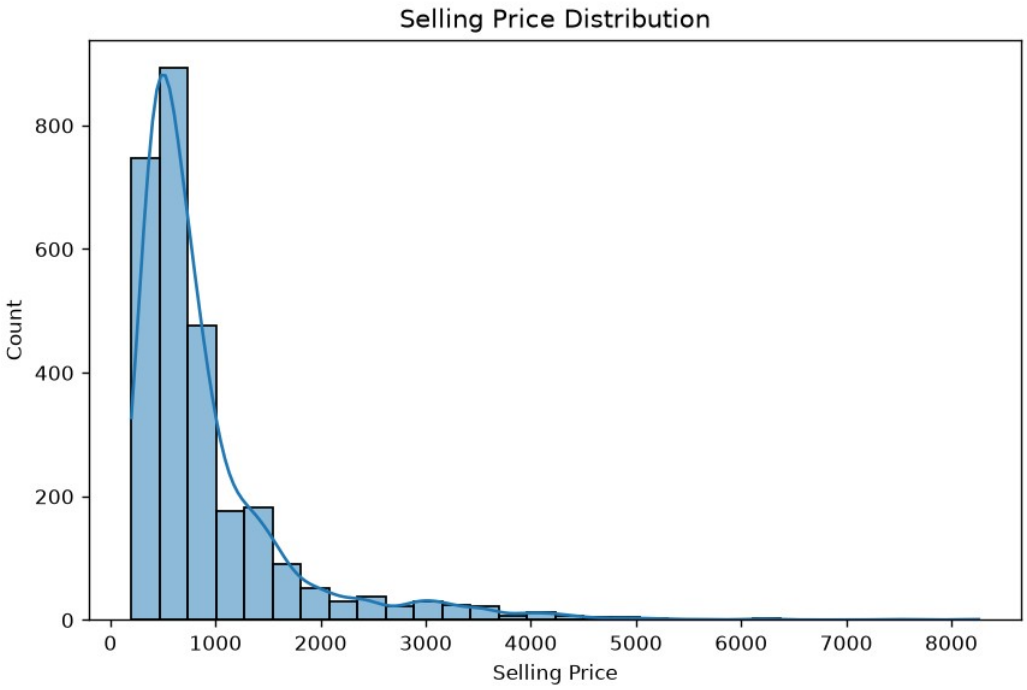
Top 10 Brands



Observation

The dataset contains several popular brands with varying product counts.

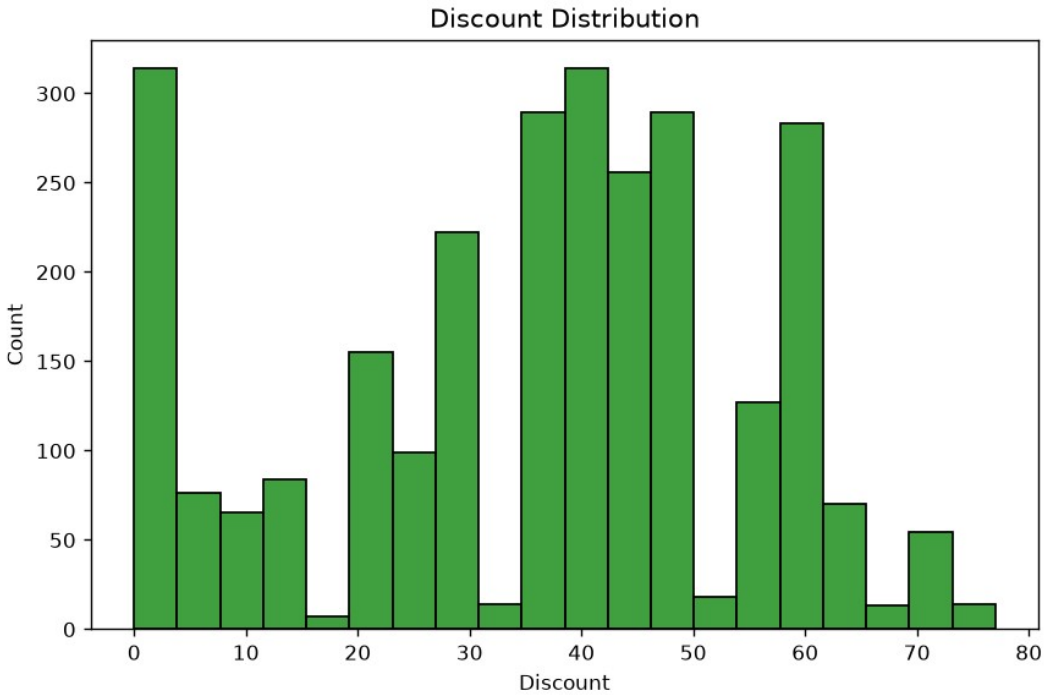
Selling Price Distribution



Observation

Most products fall within a specific price range.

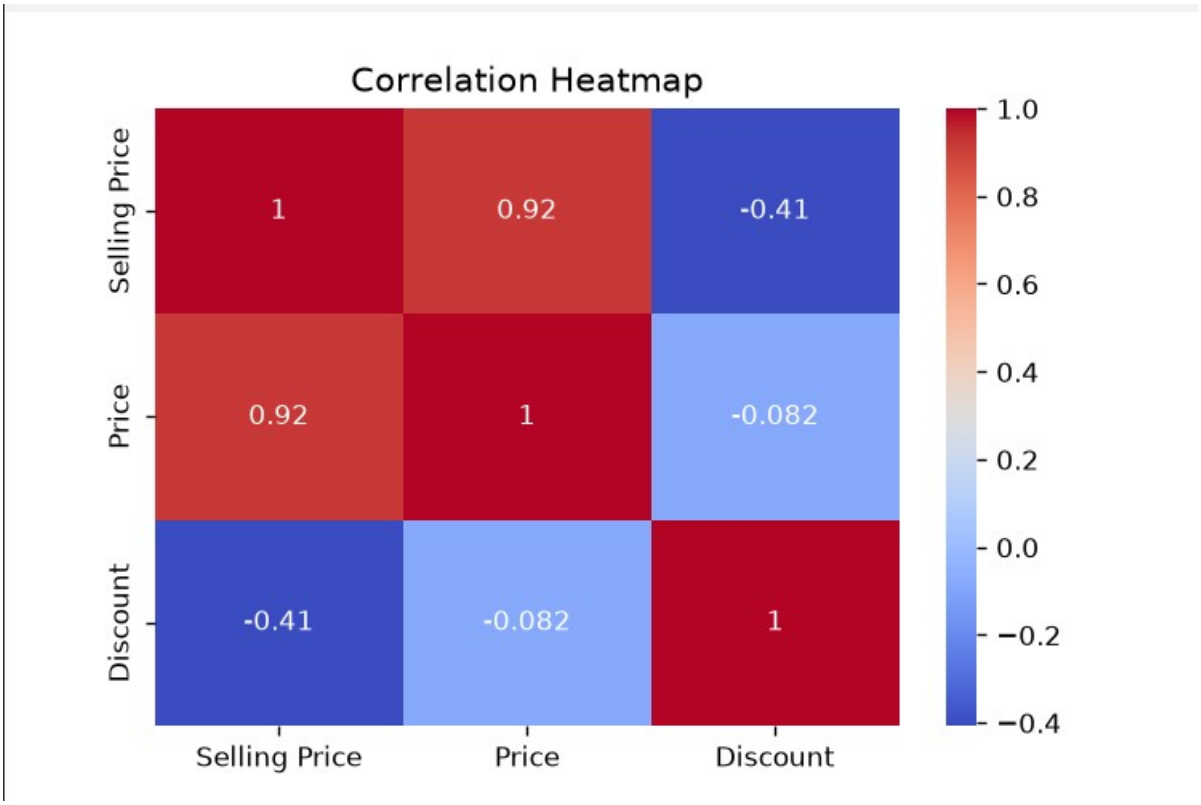
Discount Distribution



Observation

Many products are sold with discounts ranging from moderate to high percentages.

Correlation Heatmap



Observation

Selling Price and Original Price show a positive correlation.

5. Findings

- Dataset contains 4,950 T-shirt products.
- No missing values were found.
- Duplicate records were checked and handled.
- Outliers were identified using boxplots.
- Discounts significantly affect selling prices.
- Some brands offer products at premium prices.

6. Conclusion

The Myntra T-Shirt Dataset was successfully preprocessed and analyzed. Data cleaning involved checking missing values, duplicates, and data types. Exploratory Data Analysis provided insights into pricing patterns, discount trends, and brand distributions. The dataset is suitable for further machine learning and business analytics applications.